

Lesson 18: Ship It

You built an app -- now finish it

You have come from main() to Firebase. Before your app reaches real users, run through this final checklist. Each step is small, but skipping them is how broken apps get published.

Step 1: Read the error messages

When something breaks, Flutter prints an error in red. Beginners often panic and skip it -- but the message usually tells you the fix. Read it slowly:

- Null check operator used on a null value -- you used ! on something that was null. Check where the value comes from.
- setState() called after dispose() -- an async task finished after the screen was closed. Check if the widget is still on screen before calling setState().
- A long widget tree dump -- scroll to the **first** red line. That is the real error; the rest is context.

If you are stuck for more than 20 minutes, copy the first red line into a search engine. Someone has hit the same error before.

Step 2: Run flutter doctor

Open a terminal and run:

```
flutter doctor
```

It checks your whole setup -- Flutter SDK, Android tools, connected devices -- and tells you exactly what is missing, with commands to fix it. Run it any time your build fails for no clear reason.

Step 3: Build the release version

Until now you ran debug builds (fast to build, full of helper tools). A release build is what users get: smaller, faster, no debug helpers.

For Android, one command builds an installable file:

```
flutter build apk --release
```

You will find the file at build/app/outputs/flutter-apk/app-release.apk. Install it on a real phone and tap through everything -- release builds sometimes behave differently from debug ones.

Step 4: The pre-publish checklist

- Test on a real device, not just the emulator. Slow phones show problems fast phones hide.
- Check both themes if your app has dark mode. Every screen, both themes.
- Try bad input. Empty forms, huge text, no internet -- your app should handle these calmly, not crash.
- Lock your Firebase rules so users can only read and write their own data (Lesson 16).
- Remove test code -- print statements, dummy users, hard-coded test data.

Where to learn next

This course gave you the full path from zero to a shipped app. To go deeper:

- flutter.dev -- the official docs, with guides for every widget.
- Dart language tour -- learn the language behind Flutter properly.
- Build a real project -- a todo app with Firebase, a weather app with an API, a chat app. Real projects teach what tutorials cannot.

You now know enough to build and publish a real Flutter app. The rest is practice.

Key terms

- Debug build -- the slow, helper-filled version you run while developing.
- Release build -- the small, fast version users install.
- APK -- the installable app file for Android.
- flutter doctor -- the command that checks your Flutter setup and reports problems.
- Security rules -- Firebase settings that control who can read and write your data.